

claude-windows / c7679509-ee2b-4443-b240-0b3b1b1a7291

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c7679509-ee2b-4443-b240-0b3b1b1a7291\subagents\agent-a9ce9fd6661c86dc0.jsonl`  
- SHA-256: `b261ac15f2e0955bc4d2c969f713d088999286bdfa48b611c8b0d44b2586a48d`  
- Source modified: `2026-06-21T17:45:45+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `subagents`  
- session_id: `c7679509-ee2b-4443-b240-0b3b1b1a7291`
```

Transcript

1. user (2026-06-21T17:42:16.762Z)

Adversarially review PR P1b (PACKAGING_PLAN.md) in the git worktree C:/Users/avidu/Projects/bhi-wt-p1b. It bundles the khelsutra web SPA into the engine wheel and serves it single-origin.

Read in that worktree:

```
- scripts/bundle_ui.sh (source-agnostic bundler: build a khelsutra checkout now / unpack a dist tarball later; writes backend/ui/ + ui_version.json)  
- backend/main.py: the SPA serving block (search "_BUNDLED_UI_DIR", "_UI_DIR", "/assets", "serve_index", "_bundled_ui_version", "api_version")  
- backend/ui/ui_version.json (the committed snapshot's provenance)  
- tests/test_ui_bundle.py  
- .gitattributes (the `backend/ui/** -text` entry)
```

Already verified: the built wheel includes all backend/ui/ files (no force-include needed ? hatchling auto-includes backend/ subdirs); full suite 1295 passed/7 skipped; ruff+mypy clean; coverage 85.12%; tests assert / serves the SPA, /assets serves the bundle byte-for-byte, /api/version reports the ui, and a skew-guard asserts ui_version.json.engine_api_version == main.API_VERSION.

Adversarially check, with file:line evidence:

1. SERVING CORRECTNESS: serve_index serves _UI_DIR/index.html; /assets is mounted to _UI_DIR/assets. Does the /assets StaticFiles mount conflict with any API route or the legacy /static mount? Is the mount registered safely at import (dir exists)? When backend/ui is ABSENT (source checkout, no snapshot), does _UI_DIR fall back to the legacy static dir without crashing (the /assets mount guarded by isdir)? Could a request to /assets/../../something escape (StaticFiles traversal)?
2. ROUTE ORDER / SHADOWING: there is NO catch-all history-fallback (claim: SPA is query-param driven, not path-routed). Confirm no `{path:path}` catch-all was added that could shadow /api. Does serving index.html at "/" interfere with /api/* or /static?
3. BUNDLER ROBUSTNESS (scripts/bundle_ui.sh): set -euo pipefail correct? The option-3 tarball/URL branch (curl|tar) ? any injection or unsafe extraction (tar path traversal)? The content_hash + engine_api_version extraction (grep API_VERSION) ? robust? Does `cp -r "\$dist"/. ` correctly populate a cleaned dest? Does it run in CI (it should NOT ? the snapshot is committed; CI never bundles)?
4. SKEW GUARD: is the contract test's assertion (ui_version.json.engine_api_version == main.API_VERSION) a genuine guard ? i.e., if someone bumps API_VERSION without re-bundling, does it fail? Any way it passes spuriously?
5. CRLF / BINARY: does `.gitattributes: backend/ui/\*\* -text` correctly prevent autocrlf from mangling the committed JS/CSS/HTML so served bytes match vite output? Any file under backend/ui that should NOT be -text?
6. LICENSING/SIZE: the committed bundle (~700KB) includes minified third-party JS (react etc.) + OFL Noto fonts ? is committing built JS into the MIT engine repo a concern the PR should NOTE (the python license-scan lane does NOT scan JS deps)? Not necessarily a blocker ? flag if it warrants a note.
7. /api/version ROBUSTNESS: _bundled_ui_version reads backend/ui/ui_version.json; returns None on absence/parse-error (so /api/version still 200s without a bundle)? json imported?

Return ONLY: findings as "[SEVERITY blocker|major|minor|nit] <file:line> ? <issue> ? <fix>", then a one-line verdict. Be precise; don't invent issues. If clean, say so.

2. assistant (2026-06-21T17:42:19.443Z)

I'll adversarially review PR P1b. Let me start by pulling the worktree fresh and reading the key files.

3. user (2026-06-21T17:42:22.813Z)

Already up to date.

---BRANCH---

feat/packaging-p1b-bundle-ui

---HEAD---

be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities, /api/videos, /api/reels (#300)

cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)

c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)

4. user (2026-06-21T17:42:22.823Z)

```
1      #!/usr/bin/env bash
2      # Bundle the KhelSutra web SPA into the engine so a downloaded engine serves the real UI
3      # single-origin (PACKAGING_PLAN.md P1b).
4      #
5      # OPTION 1 (now, owner-chosen): build the SPA from a khelsutra checkout and vendor the
built `dist/`
6      # into `backend/ui/`, which is COMMITTED ? so the wheel + CI build fully
offline/reproducibly. Refresh
7      # the snapshot by re-running this on an SPA release; commit the diff.
8      #
9      # OPTION 3 (designed-for, future): point --from at a published dist tarball/URL (a
GitHub Release
10     # artifact) to vendor a pre-built SPA without a local khelsutra checkout or Node.
Already wired below.
11     #
12     # Usage:
13     #   scripts/bundle_ui.sh [SOURCE]
14     #   SOURCE = a khelsutra checkout dir (default: $KHELSUTRA_DIR or ../khelsutra)
[builds it]
15     #           | a path/URL ending .tgz/.tar.gz
[unpacks it]
16     #   BUILD=0 scripts/bundle_ui.sh <khelsutra-dir>    # skip npm; reuse an existing
web/dist
17     set -euo pipefail
18
19     ENGINE_ROOT="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
20     UI_DEST="$ENGINE_ROOT/backend/ui"
21     SOURCE="${1:-${KHELSUTRA_DIR:-$ENGINE_ROOT/../../khelsutra}}"
22
23     stage_dist() { # populates $1 (a dist dir) into a clean $UI_DEST
24         local dist="$1"
25         [ -f "$dist/index.html" ] || { echo "ERROR: no index.html in $dist ? not a built SPA
dist" >&2; exit 1; }
26         rm -rf "$UI_DEST"
27         mkdir -p "$UI_DEST"
28         cp -r "$dist"/. "$UI_DEST"/
29     }
30
31     SPA_REPO="unknown"
32     SPA_COMMIT="unknown"
33
34     case "$SOURCE" in
```

```

35     *.tgz|*.tar.gz|http://*|https://*)
36     # OPTION 3: a published dist artifact.
37     tmp="$(mktemp -d)"; trap 'rm -rf "$tmp"' EXIT
38     if [ [ "$SOURCE" == http*://* ] ]; then
39         echo "[bundle_ui] fetching dist artifact: $SOURCE"
40         curl -fsSL "$SOURCE" -o "$tmp/dist.tgz"; SRC_TGZ="$tmp/dist.tgz";
SPA_REPO="$SOURCE"
41     else
42         SRC_TGZ="$SOURCE"; SPA_REPO="$SOURCE"
43     fi
44     tar -xzf "$SRC_TGZ" -C "$tmp"
45     # accept either dist/* at the root or a top-level dir containing index.html
46     if [ -f "$tmp/index.html" ]; then DIST="$tmp"; else DIST="$(dirname "$(find "$tmp"
-name index.html | head -1)"); fi
47     stage_dist "$DIST"
48     ;;
49 *)
50     # OPTION 1: a khelsutra checkout ? build it.
51     WEB="$SOURCE/web"
52     [ -d "$WEB" ] || { echo "ERROR: $WEB not found (pass a khelsutra checkout dir or a
dist tarball)" >&2; exit 1; }
53     SPA_REPO="$(git -C "$SOURCE" config --get remote.origin.url 2>/dev/null || echo
"$SOURCE")"
54     SPA_COMMIT="$(git -C "$SOURCE" rev-parse HEAD 2>/dev/null || echo unknown)"
55     if [ "${BUILD:-1}" = "1" ]; then
56         echo "[bundle_ui] building SPA in $WEB (npm ci && npm run build)"
57         ( cd "$WEB" && npm ci && npm run build )
58     else
59         echo "[bundle_ui] BUILD=0 ? reusing existing $WEB/dist"
60     fi
61     stage_dist "$WEB/dist"
62     ;;
63 esac
64
65     # Provenance + the SPA<->engine contract version (the engine's current API_VERSION at
bundle time).
66     ENGINE_API_VERSION="$(grep -E '^API_VERSION\s*=' "$ENGINE_ROOT/backend/main.py" | grep
-oE '[0-9]+' | head -1)"
67     CONTENT_HASH="$(find "$UI_DEST" -type f ! -name ui_version.json | LC_ALL=C sort | xargs
sha256sum | sha256sum | cut -d' ' -f1)"
68     BUILT_AT="$(date -u +%Y-%m-%dT%H:%M:%SZ 2>/dev/null || echo unknown)"
69
70     cat > "$UI_DEST/ui_version.json" <<JSON
71     {
72         "spa_repo": "$SPA_REPO",
73         "spa_commit": "$SPA_COMMIT",
74         "engine_api_version": ${ENGINE_API_VERSION:-null},
75         "content_hash": "$CONTENT_HASH",
76         "built_at": "$BUILT_AT"
77     }
78     JSON
79
80     echo "[bundle_ui] vendored SPA -> $UI_DEST (spa_commit=$SPA_COMMIT,
engine_api_version=$ENGINE_API_VERSION)"
81     echo "[bundle_ui] files:"; find "$UI_DEST" -type f | sed "s|$UI_DEST/||" | sort
82

```

5. assistant (2026-06-21T17:42:26.087Z)

Now let me read the SPA serving block in main.py and the other key files.

6. user (2026-06-21T17:42:27.941Z)

```

34:from fastapi.staticfiles import StaticFiles
127:# arbitrary directory finds the shipped assets instead of creating a stray

```

```

./backend/api/static.
136:app.mount("/static", StaticFiles(directory=_STATIC_DIR), name="static")
141:# is needed; it is query-param driven (?video=?), so serving index.html at `` + /assets is
enough
143:_BUNDLED_UI_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "ui")
144:_UI_DIR = _BUNDLED_UI_DIR if os.path.isfile(os.path.join(_BUNDLED_UI_DIR, "index.html"))
else _STATIC_DIR
145:_UI_ASSETS_DIR = os.path.join(_UI_DIR, "assets")
147:    app.mount("/assets", StaticFiles(directory=_UI_ASSETS_DIR), name="assets")
151:def serve_index():
152:    index_path = os.path.join(_UI_DIR, "index.html")
159:def _bundled_ui_version() -> Optional[Dict[str, Any]]:
161:    Includes `engine_api_version` ? the API_VERSION the snapshot was built against (skew
guard)."""
162:    path = os.path.join(_BUNDLED_UI_DIR, "ui_version.json")
260:API_VERSION = 1
272:def api_version():
274:    return {"engine_version": _engine_version(), "api_version": API_VERSION, "ui":
_bundled_ui_version()}
295:        "api_version": API_VERSION,

```

7. user (2026-06-21T17:42:28.283Z)

```

1      {
2      "spa_repo": "https://github.com/avidullu/khelsutra.git",
3      "spa_commit": "46e0244b0c862a4a9d845b8947843b5dfa1f2d9f",
4      "engine_api_version": 1,
5      "content_hash": "fad6253629831592b11dc59a1b4546af8a1fa96cc13c498a9255b47dcf0d04ff",
6      "built_at": "2026-06-21T17:35:18Z"
7      }
8

```

8. user (2026-06-21T17:42:29.220Z)

```

1      """The bundled SPA is served single-origin + the SPA<->engine contract
(PACKAGING_PLAN.md P1b)."""
2
3      import json
4      import os
5
6      import pytest
7
8      pytest.importorskip("httpx")
9      from fastapi.testclient import TestClient
10
11     import backend.main as m
12
13     client = TestClient(m.app)
14     UI_DIR = os.path.join(os.path.dirname(os.path.abspath(m.__file__)), "ui")
15
16
17     def test_bundled_ui_snapshot_is_committed():
18         assert os.path.isfile(os.path.join(UI_DIR, "index.html")), (
19             "backend/ui/index.html missing ? the SPA snapshot must be committed (run
scripts/bundle_ui.sh).")
20         assert os.path.isfile(os.path.join(UI_DIR, "ui_version.json"))
21
22
23     def test_serves_spa_at_root():
24         r = client.get("/")
25         assert r.status_code == 200
26         # The bundled index.html references its single-origin /assets bundle (Vite
base='/').
27         assert "/assets/" in r.text
28

```

```

29
30     def test_serves_spa_assets_byte_for_byte():
31         assets = os.path.join(UI_DIR, "assets")
32         js = next(f for f in os.listdir(assets) if f.endswith(".js"))
33         r = client.get(f"/assets/{js}")
34         assert r.status_code == 200
35         with open(os.path.join(assets, js), "rb") as f:
36             assert r.content == f.read() # served verbatim (mimetype varies by OS, so
compare bytes)
37
38
39     def test_api_version_reports_bundled_ui():
40         body = client.get("/api/version").json()
41         assert body["api_version"] == m.API_VERSION
42         assert body["ui"] is not None
43         assert body["ui"]["spa_commit"] # provenance recorded
44
45
46     def test_spa_engine_contract_version_matches():
47         """Skew guard: the committed snapshot records the engine API_VERSION it was built
against. If
48         someone bumps API_VERSION without re-running scripts/bundle_ui.sh, this FAILS ?
forcing a
49         re-bundle so the shipped SPA can't silently target a stale API contract."""
50         with open(os.path.join(UI_DIR, "ui_version.json"), encoding="utf-8") as f:
51             uiv = json.load(f)
52         assert uiv["engine_api_version"] == m.API_VERSION, (
53             f"bundled SPA built against engine api_version {uiv['engine_api_version']} but
engine is now "
54             f"{m.API_VERSION} ? re-run scripts/bundle_ui.sh and commit backend/ui/.")
55

```

9. user (2026-06-21T17:42:29.373Z)

```

1     # Golden Regression Fixtures (M1) ? pin LF line endings so the committed fixtures are
2     # byte-stable across OSES despite core.autocrlf. The characterization gate parse-
compares
3     # (never byte-compares) JSON, but a stable on-disk form keeps `git diff` reviewable and
the
4     # fixtures reproducible. See docs/GOLDEN_REGRESSION_FIXTURES.md §4d/§4e.
5     eval_baselines/fixtures/** text eol=lf
6
7     # Shell scripts must stay LF so they run on Linux boxes (the DigitalOcean
8     # bootstrap) regardless of a contributor's core.autocrlf. A CRLF shebang
9     # (`#!/usr/bin/env bash\r`) fails on Linux. See deploy/digitalocean/.
10    *.sh text eol=lf
11
12    # Vendored SPA build output (PACKAGING_PLAN.md P1b): commit byte-exact with NO EOL
conversion, so the
13    # served assets match vite's output cross-platform regardless of core.autocrlf. Refresh
via
14    # scripts/bundle_ui.sh. (woff2 are auto-detected binary; this pins the
.js/.css/.html/.json too.)
15    backend/ui/** -text
16

```

10. assistant (2026-06-21T17:42:32.327Z)

Now let me read the full serving block in main.py around lines 120-300 to get the exact context.

11. user (2026-06-21T17:42:33.260Z)

```

118         if env and env.strip():
119             return env.strip()

```

```

120     sec = getattr(config, "security", None)
121     if getattr(sec, "edge_auth_enabled", False):
122         return "0.0.0.0" # behind a Cloudflare-Access-style edge gate (M9) ? bind all
interfaces
123         return "127.0.0.1"
124
125     # Serves static frontend files directly (now under api/ per approved structure).
126     # PACKAGING_PLAN.md P0a: resolve PACKAGE-relative (not CWD-relative) so launching from
an
127     # arbitrary directory finds the shipped assets instead of creating a stray
./backend/api/static.
128     # (The legacy in-engine dashboard; the real SPA gets bundled here in P1a.)
129     _STATIC_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "api", "static")
130     try:
131         os.makedirs(_STATIC_DIR, exist_ok=True)
132     except OSError:
133         # The dir ships with the package (a no-op create); guard against a read-only install
134         # tree (system-installed Python) since exist_ok suppresses FileExistsError, not
PermissionError.
135         pass
136     app.mount("/static", StaticFiles(directory=_STATIC_DIR), name="static")
137
138     # P1b: serve the bundled KhelSutra SPA single-origin. `backend/ui/` is the committed
snapshot
139     # (produced by scripts/bundle_ui.sh); fall back to the legacy in-engine dashboard only
if it's
140     # absent (a source checkout with no vendored UI). The SPA calls same-origin `/api`, so
no SPA change
141     # is needed; it is query-param driven (?video=?), so serving index.html at `/` + /assets
is enough
142     # (no client-side path routing ? no history-fallback catch-all, which would also be a
route-order hazard).
143     _BUNDLED_UI_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "ui")
144     _UI_DIR = _BUNDLED_UI_DIR if os.path.isfile(os.path.join(_BUNDLED_UI_DIR, "index.html"))
else _STATIC_DIR
145     _UI_ASSETS_DIR = os.path.join(_UI_DIR, "assets")
146     if os.path.isdir(_UI_ASSETS_DIR):
147         app.mount("/assets", StaticFiles(directory=_UI_ASSETS_DIR), name="assets")
148
149
150     @app.get("/", response_class=HTMLResponse)
151     def serve_index():
152         index_path = os.path.join(_UI_DIR, "index.html")
153         if os.path.exists(index_path):
154             with open(index_path, "r", encoding="utf-8") as f:
155                 return f.read()
156         return "<h3>Sports Highlight Indexer Server is Running. (No bundled UI found.)</h3>"
157
158
159     def _bundled_ui_version() -> Optional[Dict[str, Any]]:
160         """Provenance of the vendored SPA (backend/ui/ui_version.json), or None if no UI is
bundled.
161         Includes `engine_api_version` ? the API_VERSION the snapshot was built against (skew
guard)."""
162         path = os.path.join(_BUNDLED_UI_DIR, "ui_version.json")
163         try:
164             with open(path, "r", encoding="utf-8") as f:
165                 return json.load(f)
166         except (OSError, ValueError):
167             return None
168
169     # Global variables initialized at startup
170     db_instance: Optional[Database] = None
171     global_config: Optional[AppConfig] = None # now typed (Pydantic model)
172     # P0e: the single-instance lock handle. Kept module-global for the process lifetime ? if

```

```

it were
173     # GC'd/closed the OS would release the lock. None until main() acquires it (server/CLI
init).
174     _instance_lock: Optional[Any] = None
175
176     # Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md
P3)
177     # now lives in backend.api.job_worker. Re-exported here so the names stay addressable on
178     # `backend.main` (`JobWaiting`, `_drain_due_jobs`, `_arm_wake_timer`, `_get_executor`,
179     # `_run_claimed_job`, `_MAX_DRAIN_WAKE_SEC`, ...). The worker resolves these seams
THROUGH
180     # this module object at call-time, so `monkeypatch.setattr(backend.main, ...)` still
181     # intercepts the inter-function calls (a parked job arms the PATCHED `_arm_wake_timer`,
182     # so no real daemon timer leaks in tests). Behavior is unchanged by the extraction.
183     from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable
on backend.main)
184         JobWaiting,
185         _arm_wake_timer,
186         _drain_due_jobs,
187         _get_executor,
188         _job_to_request,
189         _MAX_DRAIN_WAKE_SEC,
190         _maybe_record_job_cost,
191         _run_claimed_job,
192         _schedule_next_drain_if_waiting,
193     )
194
195     # Legacy thin wrapper for any remaining direct calls during transition.
196     # New code should import load_app_config / AppConfig from backend.config directly.
197     def load_config(config_path: str = "config.json") -> Dict[str, Any]:
198         cfg = load_app_config(config_path)
199         return cfg.model_dump(mode="json")
200
201     # --- API Models ---
202     class ProcessRequest(BaseModel):
203         video_path: str
204         player_pool: Optional[List[str]] = None
205         max_frames: Optional[int] = None
206         segmenter: Optional[str] = None
207
208     class QueryRequest(BaseModel):
209         video_id: str
210         server: Optional[str] = None
211         receiver: Optional[str] = None
212         duration_min: Optional[float] = None
213         event_type: Optional[str] = None
214
215     class CompileRequest(BaseModel):
216         video_id: str
217         segment_ids: List[int]
218         output_filename: str
219
220     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
221         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
222
223         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
224         fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark)
and
225         keep the prior good results intact ? a failed/empty re-run never destroys prior
data.
226         """
227         if segments:
228             db_instance.prune_segments(video_id, play_upto=play_wm, cand_upto=cand_wm)
229         else:
230             db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)

```

```

231
232
233     # --- API Endpoints ---
234     @app.get("/api/config")
235     def get_config():
236         # P0b: this endpoint is unauthenticated, so never serialize a credential-shaped
field. Today
237         # AppConfig holds no secrets (the Gemini key is env-only; cf_access_aud is a public
id), so this
238         # is byte-identical for the real fields ? but it fails-closed if a secret is ever
added (guarded
239         # by tests/test_redact.py + the /api/config redaction test).
240         if global_config is None:
241             return {}
242         return redact_secrets(global_config.model_dump(mode="json"))
243
244     @app.get("/api/health")
245     def health():
246         """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
247         (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
248         confirm the engine is reachable and which detector is wired. Never raises."""
249         sport = getattr(global_config, "sport", None)
250         indexing = getattr(global_config, "indexing", None)
251         return {
252             "status": "ok",
253             "sport": sport,
254             "default_segmenter": getattr(indexing, "default_segmenter", None),
255         }
256
257
258     # P1: the bundled SPA pins this ? the contract version of the /api surface. BUMP when an
existing
259     # endpoint's request/response shape changes incompatibly (purely additive endpoints need
no bump).
260     API_VERSION = 1
261
262
263     def _engine_version() -> str:
264         """Installed package version (wheel/editable); 'unknown' if not importable as a
distribution."""
265         try:
266             return _pkg_version("badminton-highlight-indexer")
267         except PackageNotFoundError:
268             return "unknown"
269
270
271     @app.get("/api/version")
272     def api_version():
273         """Version handshake so a bundled SPA can detect engine skew (PACKAGING_PLAN.md
P1)."""
274         return {"engine_version": _engine_version(), "api_version": API_VERSION, "ui":
_bundled_ui_version()}
275
276
277     @app.get("/api/capabilities")
278     def capabilities():
279         """Honest probe of what THIS box can do ? the first-run wizard renders the compute
step from
280         it (PACKAGING_PLAN.md P1). No secrets: reports only whether a Gemini key is PRESENT,
never its
281         value. Best-effort + never raises (the wizard degrades gracefully)."""
282         indexing = getattr(global_config, "indexing", None)
283         deployment = getattr(global_config, "deployment", None)
284         ffmpeg = shutil.which("ffmpeg")
285         ffmpegprobe = shutil.which("ffmpegprobe")

```

```

286         # WASB weights are best-effort: env first (matches native_wasb_runner), then config.
287         wasb = getattr(indexing, "wasb", None)
288         weights_path = os.environ.get("WASB_WEIGHTS") or getattr(wasb, "weights_path", "")
289         or ""
290         # Cheap GPU heuristic: nvidia-smi on PATH (matches scripts/doctor.py). Deliberately
291         NOT
292         # torch.cuda.is_available() ? torch is absent in the LIGHT tier (so it would false-
293         negative on a
294         # real GPU box) and importing it would add seconds to this probe. The detector
295         healthcheck is the
296         # authority on whether GPU inference actually works; this is just the wizard's
297         first-order signal.
298         nvidia_smi = shutil.which("nvidia-smi")
299         return {
300             "api_version": API_VERSION,
301             "engine_version": _engine_version(),
302             "segmenters": sorted(segmenter_registry.list_available().keys()),
303             "default_segmenter": getattr(indexing, "default_segmenter", None),
304             "ffmpeg": {"ffmpeg": bool(ffmpeg), "ffprobe": bool(ffprobe),
305                        "ffmpeg_path": ffmpeg, "ffprobe_path": ffprobe},
306             "gpu": {"nvidia_smi_present": bool(nvidia_smi)},
307             "gemini_key_present": bool(os.environ.get("GEMINI_API_KEY")),

```

12. assistant (2026-06-21T17:42:36.917Z)

Now I have the serving block. Let me verify a few things: that there's no catch-all route, that `json`, `Optional`, `Dict`, `Any` are imported, that `/assets` mount doesn't conflict with any API route, and check the actual contents of backend/ui to assess the licensing/size concern.

13. user (2026-06-21T17:42:39.490Z)

```

136:app.mount("/static", StaticFiles(directory=_STATIC_DIR), name="static")
147:    app.mount("/assets", StaticFiles(directory=_UI_ASSETS_DIR), name="assets")
150:@app.get("/", response_class=HTMLResponse)
234:@app.get("/api/config")
244:@app.get("/api/health")
271:@app.get("/api/version")
277:@app.get("/api/capabilities")
311:@app.get("/api/videos")
325:@app.get("/api/reels")
356:@app.get("/api/reels/{filename}")
496:@app.post("/api/process", status_code=202)
547:@app.get("/api/jobs/{job_id}/cost")
562:@app.get("/api/videos/{video_id}/cost")
573:@app.get("/api/jobs/{job_id}")
606:@app.get("/api/highlights/{video_id}/{filename}")
630:@app.post("/api/query")
641:@app.post("/api/compile")

```

14. user (2026-06-21T17:42:40.611Z)

```

1:import os
2:import sys
3:import argparse
4:import json
5:import logging
6:import shutil
7:import uuid
8:from importlib.metadata import PackageNotFoundError, version as _pkg_version
9:from urllib.parse import quote
10:import uvicorn
11:from dotenv import load_dotenv
17:from backend.utils.app_home import env_file_to_load, resolve_output_dir
28:from fastapi import FastAPI, HTTPException, Request

```

```

29:from fastapi.middleware.cors import CORSMiddleware
30:from starlette.middleware.trustedhost import TrustedHostMiddleware
31:from pydantic import BaseModel
32:from typing import List, Dict, Any, Optional
34:from fastapi.staticfiles import StaticFiles
35:from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
36:from backend.pipeline.validators.base import registry
37:from backend.infrastructure.database import Database
38:from backend.pipeline.segmenters.base import segmenter_registry
39:from backend.pipeline.stitcher.compiler import VideoCompiler
40:from backend.utils.hashing import compute_video_id # canonical file-identity hashing
41:from backend.utils.paths import resolve_under_root, safe_output_filename, validate_input_path
42:from backend.utils.redact import redact_secrets # P0b: mask secret-shaped fields in
/api/config
43:from backend.utils.single_instance import acquire_lock # P0e: one instance per database
44:from backend.config import ( # new typed config
51:from backend.results import Failure # standardized error/result contract
52:from backend.reporting import emit_indexer_report
53:from backend import storage # M2: URI-aware input acquisition (local = identity passthrough)
54:from backend import billing # M8: per-job cost ledger (USD internal / INR display)
55:from backend.api import auth as edge_auth # M9: Cloudflare Access edge-auth (default-OFF)
60:def _cors_origins_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
83:def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
110:def _resolve_bind_host(config: Optional[AppConfig] = None, cli_host: Optional[str] = None)
-> str:
159:def _bundled_ui_version() -> Optional[Dict[str, Any]]:
170:db_instance: Optional[Database] = None
171:global_config: Optional[AppConfig] = None # now typed (Pydantic model)
174:_instance_lock: Optional[Any] = None
183:from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable on
backend.main)
197:def load_config(config_path: str = "config.json") -> Dict[str, Any]:
204:    player_pool: Optional[List[str]] = None
205:    max_frames: Optional[int] = None
206:    segmenter: Optional[str] = None
210:    server: Optional[str] = None
211:    receiver: Optional[str] = None
212:    duration_min: Optional[float] = None
213:    event_type: Optional[str] = None
312:def list_videos(limit: Optional[int] = None):

```

[Showing results with pagination = limit: 50]

15. assistant (2026-06-21T17:42:44.087Z)

Confirmed: `json` imported (line 4), `Optional/Dict/Any` imported (line 32). No `{path:path}` catch-all exists ? only specific `/api/\*` routes and `/` plus the two static mounts. Let me inspect the bundled UI contents for the licensing/size concern and verify the assets dir structure.

16. user (2026-06-21T17:42:46.702Z)

```

=== backend/ui tree ===
backend/ui/assets/index-FJh_kElx.css
backend/ui/assets/index-ZjpoX6Fg.js
backend/ui/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
backend/ui/assets/noto-sans-devanagari-latin-ext-wght-normal-CLKupaxV.woff2
backend/ui/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
backend/ui/assets/noto-sans-kannada-kannada-wght-normal-DuWB-P8d.woff2
backend/ui/assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
backend/ui/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
backend/ui/index.html
backend/ui/ui_version.json

=== sizes ===

```

702K backend/ui

```
=== total + per-file size ===
373799 backend/ui/assets/index-ZjpoX6Fg.js
121192 backend/ui/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
90796 backend/ui/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
27576 backend/ui/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
25384 backend/ui/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
24831 backend/ui/assets/index-FJh_kElx.css
13992 backend/ui/assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
13852 backend/ui/assets/noto-sans-devanagari-latin-ext-wght-normal-CLkupaxV.woff2
401 backend/ui/index.html
273 backend/ui/ui_version.json
```

17. assistant (2026-06-21T17:42:51.339Z)

Now let me verify a few critical claims. First, does StaticFiles protect against path traversal (it does by default via Starlette). Let me verify the skew-guard actually works by checking the content_hash relationship, and verify the `/assets` mount vs `/api` route shadowing. Let me also check whether the index.html references `/assets/` and confirm Vite base. Let me also verify the git attributes are actually applied (check for committed CRLF) and confirm the content_hash in ui_version.json ? the bundler EXCLUDES ui_version.json from the hash but the test only checks engine_api_version, not content_hash.

18. user (2026-06-21T17:42:52.302Z)

```
1      <!doctype html>
2      <html lang="en">
3          <head>
4              <meta charset="UTF-8" />
5              <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6              <title>KhelSutra Studio</title>
7              <script type="module" crossorigin src="/assets/index-ZjpoX6Fg.js"></script>
8              <link rel="stylesheet" crossorigin href="/assets/index-FJh_kElx.css">
9          </head>
10         <body>
11             <div id="root"></div>
12         </body>
13     </html>
14
```

19. user (2026-06-21T17:42:55.088Z)

```
=== git check-attr on UI files ===
backend/ui/index.html: text: unset
backend/ui/assets/index-ZjpoX6Fg.js: text: unset
backend/ui/ui_version.json: text: unset

=== CRLF check (look for \r) ===
fatal: path 'backend/ui/index.html' exists on disk, but not in 'HEAD'
/dev/stdin: empty

=== is ui_version.json -text too? (should it be eol-converted?) ===
eol: text: unspecified
backend/ui/ui_version.json: text: unset
```

20. assistant (2026-06-21T17:42:58.058Z)

The files exist on disk but not in HEAD ? they may be staged but not committed. Let me check git status of the worktree to confirm whether these files are actually committed (this matters: the PR claims the snapshot is committed).

21. user (2026-06-21T17:43:00.982Z)

```

=== branch / HEAD ===
feat/packaging-p1b-bundle-ui
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities,
/api/videos, /api/reels (#300)

=== status ===
M .gitattributes
M backend/main.py
?? backend/ui/
?? scripts/bundle_ui.sh
?? tests/test_ui_bundle.py

=== are UI files tracked at HEAD? ===
NOT IN HEAD

=== commits on this branch vs master ===

```

22. assistant (2026-06-21T17:43:08.413Z)

Important finding: the PR is **entirely uncommitted working-tree changes** ? nothing is committed on this branch yet. `backend/ui/` is untracked (`??`), `.gitattributes` and `backend/main.py` are modified but not committed. This means:

1. The `.gitattributes: backend/ui/** -text` rule is **not yet effective for committed files** ? `git check-attr` shows `text: unset` (the `-text` rule IS being read, good), but since the files are untracked, git hasn't yet normalized/stored them. The byte-for-byte test passes now only because it reads working-tree files directly. The real question is whether, once committed, the bytes survive. Since `.gitattributes` is in the working tree, `git add` will apply `-text` (no conversion). That's correct ? but worth confirming the rule actually matches.

`git check-attr -a` only showed `text: unset` ? let me confirm the `-text` is what produces that and that woff2/binary files are handled. Let me also verify the skew guard logic more deeply and check the bundler's hash robustness.

23. user (2026-06-21T17:43:15.011Z)

```

=== check-attr -text specifically ===
backend/ui/index.html: text: unset
backend/ui/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2: text: unset

=== does -text resolve to 'unset'? (yes = correct, no conversion) ===
'text: unset' means -text applied -> no EOL conversion. Good.

=== Simulate: would CRLF be introduced on commit+checkout? core.autocrlf setting ===
true

=== recompute content_hash the way bundle_ui.sh does, compare to ui_version.json ===
recomputed: ac2a3d672c016d57c3e237f17b9251b22a0c0a084d488906b586aeecf4a666a3
committed:  fad6253629831592b11dc59a1b4546af8a1fa96cc13c498a9255b47dcf0d04ff

```

24. assistant (2026-06-21T17:43:23.767Z)

Two important findings here:

1. **`core.autocrlf=true`** ? this is exactly the dangerous setting the `-text` attribute defends against. The `-text` rule correctly resolves (`text: unset`), so committed UI files will NOT be CRLF-mangled. Good ? the defense is correct and necessary.

2. **content_hash MISMATCH**: recomputed `ac2a3d6...` vs committed `fad6253...`. This needs investigation. The hash in `ui_version.json` does not match a fresh recompute. Possible causes: (a) the `find/sort/xargs` producing different ordering or path prefixes, or (b) the on-disk bytes differ from when the hash was generated. Let me dig in ? this is potentially a real defect (the `content_hash` is wrong, OR my recompute differs from the bundler due to the `./` path prefix). The bundler uses `find "$UI_DEST"` (absolute paths) while I used `find .` (relative). `sha256sum`

includes the filename in... no, sha256sum hashes content but outputs ``<hash> <filename>``, and the second ``sha256sum`` hashes that combined stream INCLUDING the filenames. So path prefix matters. Let me replicate exactly.

25. user (2026-06-21T17:43:28.241Z)

=== replicate bundler exactly: find \$UI_DEST (absolute) ===

```
abs-path hash: fad6253629831592b11dc59a1b4546af8a1fa96cc13c498a9255b47dcf0d04ff
committed:    fad6253629831592b11dc59a1b4546af8a1fa96cc13c498a9255b47dcf0d04ff
```

=== NOTE: hash includes absolute paths in the sha256sum stream ? so it is machine/path dependent, NOT reproducible across checkouts ===

=== show what the inner stream looks like ===

```
4c4fc6b724aad933e036c48c507fae0e393d45892d3343c99f27a3a7b1c1cbf4 */c/Users/avidu/Projects/bhi-
wt-p1b/backend/ui/assets/index-FJh_kElx.css
ebf38fa2a49c1426eec1b87bdb90188a3df94fc00cafe4601720b0428f9d43c4 */c/Users/avidu/Projects/bhi-
wt-p1b/backend/ui/assets/index-ZjpoX6Fg.js
3b3cae4d2600cb502286577fcfbc7f0caa05d9bb3c28d6699aeb56302d35e730 */c/Users/avidu/Projects/bhi-
wt-p1b/backend/ui/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
```

26. assistant (2026-06-21T17:43:34.975Z)

Confirmed: the content_hash matches when computed with absolute paths. But this reveals a **defect in `content\_hash` design**: the `find "$UI_DEST"` uses absolute paths, and the second `sha256sum` hashes a stream that **embeds those absolute paths** (``/c/Users/avidu/Projects/bhi-wt-p1b/...``). So `content_hash` is **machine/path-dependent and NOT reproducible** ? re-running the bundler on the same bytes from a different checkout path (e.g. `../khelesutra` on another machine, or CI) produces a different hash. It's recorded as provenance but is not a verifiable content fingerprint. It's also never asserted by any test, so it's effectively decorative. Minor ? but worth noting since the field implies content verifiability.

Now let me verify the StaticFiles traversal protection claim and the `/assets` vs `/api` ordering, plus confirm the full test suite / the specific claims. Let me check route matching behavior and confirm there's no `/assets`-prefixed API route.

27. user (2026-06-21T17:43:36.535Z)

```
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:34:from fastapi.staticfiles import
StaticFiles
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:136:app.mount("/static",
StaticFiles(directory=_STATIC_DIR), name="static")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:141:# is needed; it is query-param driven
(?video=?), so serving index.html at `/` + /assets is enough
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:147:    app.mount("/assets",
StaticFiles(directory=_UI_ASSETS_DIR), name="assets")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:359:    Mirrors /api/highlights' safety
(bare-name + containment checks)."""
C:\Users\avidu\Projects\bhi-
wt-p1b\backend\main.py:606:@app.get("/api/highlights/{video_id}/{filename}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:701:        download_url =
f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
C:\Users\avidu\Projects\bhi-wt-p1b\backend\ui\index.html:7:    <script type="module" crossorigin
src="/assets/index-ZjpoX6Fg.js"></script>
C:\Users\avidu\Projects\bhi-wt-p1b\backend\ui\index.html:8:    <link rel="stylesheet"
crossorigin href="/assets/index-FJh_kElx.css">
C:\Users\avidu\Projects\bhi-wt-p1b\backend\api\static\app.js:392:        const dlUrl =
`/api/highlights/${currentVideoId}/${encodeURIComponent(outputName)}`;
C:\Users\avidu\Projects\bhi-wt-p1b\backend\ui\assets\index-FJh_kElx.css:1:[Omitted long matching
line]
```

28. user (2026-06-21T17:43:40.257Z)

=== StaticFiles traversal: Starlette normalizes & blocks ../ ? verify version ===

starlette 1.0.0

=== does any /assets path appear in routes other than the mount? ===

=== confirm legacy _STATIC_DIR has NO assets/ subdir (so fallback path: /assets NOT mounted) ===
total 48

```
drwxr-xr-x 1 avidu 197609      0 Jun 21 23:04 .  
drwxr-xr-x 1 avidu 197609      0 Jun 21 23:07 ..  
-rw-r--r-- 1 avidu 197609 16350 Jun 21 23:04 app.js  
-rw-r--r-- 1 avidu 197609  8743 Jun 21 23:04 index.html  
-rw-r--r-- 1 avidu 197609 13418 Jun 21 23:04 style.css
```

29. assistant (2026-06-21T17:43:46.631Z)

Good findings:

- No `/assets`-prefixed API route exists ? no shadowing. Starlette 1.0.0 `StaticFiles` blocks `../` traversal (it resolves paths and returns 404 for escapes). Traversal is safe.
- The legacy `\_STATIC\_DIR` (`backend/api/static/`) has NO `assets/` subdir, so in the fallback case (no bundle), `\_UI\_ASSETS\_DIR` = `backend/api/static/assets` which doesn't exist ?
`os.path.isdir` is False ? `/assets` not mounted. Confirmed graceful fallback. Good.

Now, one subtle concern about **route ordering** for the fallback case: when `backend/ui` is absent, `\_UI\_DIR` falls back to `\_STATIC\_DIR`, and `serve\_index()` serves the legacy `backend/api/static/index.html` at `/`. But the legacy `index.html` references `app.js` and `style.css` ? let me check whether those are reachable. They'd be at `/static/app.js` (the legacy mount), not `/`. Let me check the legacy index.html's references and whether the fallback is actually functional. This is a real correctness question for the fallback path.

30. user (2026-06-21T17:43:48.041Z)

=== legacy index.html asset refs (does it use /static/ or /assets/?) ===

```
7:    <link rel="stylesheet" href="/static/style.css">  
145:   <script src="/static/app.js"></script>
```

31. assistant (2026-06-21T17:43:55.164Z)

The legacy fallback is fully functional: legacy `index.html` references `/static/style.css` and `/static/app.js`, which are served by the always-present `/static` mount (line 136). So when the bundle is absent, `serve\_index()` serves the legacy dashboard and its assets resolve via `/static`. The fallback doesn't crash and is coherent. Good.

Now let me verify the test suite actually runs these tests and confirm the byte-for-byte test handles the `index.html` reference correctly (it asserts `/assets/` in `r.text` ? confirmed present). Let me run the targeted test file plus confirm one edge: the `test\_serves\_spa\_assets\_byte\_for\_byte` picks the first `.js` ? but with autocrlf=true, the served bytes come from the working tree (not git), and FileResponse reads raw bytes, so it matches. That's fine in this worktree. Let me run the actual test file to confirm green.

32. user (2026-06-21T17:44:03.613Z)

```
..... [100%]  
5 passed in 5.81s
```

33. assistant (2026-06-21T17:44:11.272Z)

All 5 tests pass. Now let me verify the skew guard rigor ? can it pass spuriously? The bundler extracts `engine\_api\_version` via `grep -E '^API\_VERSION\s\*=' ... | grep -oE '[0-9]+' | head -1`. If `API\_VERSION = 1` and someone bumps to `2`, the bundler (when re-run) writes `2`. But the test compares the COMMITTED json against the live `m.API\_VERSION`. So if you bump API_VERSION in code without re-bundling, json still says old ? test fails. Good. Let me check one spurious-pass vector: what if `API\_VERSION` line has a comment with a number, or the grep picks up the wrong number. Line 258-260: there's a comment "P1:" before `API\_VERSION = 1`. The grep anchors `^API\_VERSION\s\*=` so it only matches the assignment line, then `grep -oE '[0-9]+' | head -1` takes the first integer = `1`. Robust. But let me check: what if API_VERSION were

`10`? `head -1` of `grep -oE '[0-9]+'` on `API\_VERSION = 10` gives `10`. Fine. What about `API\_VERSION: int = 1`? The anchor `^API\_VERSION\s\*=` would NOT match (there's `: int` before `=`). Let me confirm current form matches and note the brittleness.

34. user (2026-06-21T17:44:14.240Z)

=== exact API_VERSION line ===

```
161:    Includes `engine_api_version` ? the API_VERSION the snapshot was built against (skew
guard)."""
260:API_VERSION = 1
274:    return {"engine_version": _engine_version(), "api_version": API_VERSION, "ui":
_bundled_ui_version()}
295:        "api_version": API_VERSION,
```

=== test the bundler's grep against current file ===

1

=== brittleness test: would 'API_VERSION: int = 2' match? ===
NO MATCH (anchor fails on type annotation)

35. assistant (2026-06-21T17:44:21.037Z)

Confirmed the extraction works for the current `API\_VERSION = 1` form. There's a brittleness vector: if `API\_VERSION` ever gains a type annotation (`API\_VERSION: int = 1`), the grep anchor `^API\_VERSION\s\*=` would fail to match ? `ENGINE\_API\_VERSION` empty ? bundler writes `engine\_api\_version: null`. The skew test would then compare `null != 1` ? FAIL loudly. So it fails-safe (the bundler operator notices), not silently. Minor robustness note, not a guard hole.

One more spurious-pass check on the skew guard: the test reads the committed json and compares to `m.API\_VERSION`. Both are read at test time. There's no way for it to pass if they differ. The only "spurious pass" would be if both happen to be the wrong value together ? but that requires re-bundling against the bumped version, which is exactly the desired action. Guard is genuine.

Now let me check the CI question (item 3): does `bundle\_ui.sh` run in CI? Let me check CI config and confirm the snapshot is committed (not generated in CI).

36. user (2026-06-21T17:44:24.083Z)

=== CI workflows referencing bundle_ui or backend/ui ===
(no CI references ? snapshot is committed, CI never bundles. GOOD)

=== pyproject: how is backend/ui included in the wheel? force-include or auto? ===

```
3:# Build backend: hatchling. `backend/` is an intentional NAMESPACE package
4:# (no `backend/__init__.py`; mpy.ini sets explicit_package_bases for the same
5:# reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6:# auto-detection would otherwise miss the un-__init__'d `backend` root.
13:requires = ["hatchling"]
14:build-backend = "hatchling.build"
37:    # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d /
76:# Storage backends (docs/DECOUPLED_COMPUTE/). The default `local` backend needs NONE of
77:# these. Cloud SDKs are imported lazily, so `import backend.storage` works without them;
80:# MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
85:# `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
86:indexer = "backend.main:main"
88:[tool.hatch.metadata]
90:# reference*); hatchling requires opting in to allow those in metadata.
93:[tool.hatch.build.targets.wheel]
94:# `backend` is a namespace package (no __init__.py); list both import roots
95:# explicitly so an editable/wheel install exposes `import backend` and
97:packages = ["backend", "training"]
```

37. assistant (2026-06-21T17:44:30.323Z)

Important finding for the packaging claim. The comment at pyproject line 3-6 says `backend/` is a **namespace package** and "the wheel target packages are listed EXPLICITLY" because hatchling's auto-detection "would otherwise miss the un-`\_\_init\_\_`'d backend root." The PR's prior verification claims "hatchling auto-includes backend/ subdirs (no force-include needed)." Let me verify the wheel actually contains the `backend/ui/` data files ? because `packages = ["backend", "training"]` tells hatchling to include `backend/`, but `backend/ui/\*.woff2`, `\*.js`, `\*.css`, `\*.html`, `\*.json` are **non-Python data files**. Hatchling's wheel target by default includes only files under listed packages ? and it DOES include non-`.py` files under those package dirs by default (it's not setuptools `package\_data`). But this is the load-bearing claim. Let me actually build the wheel and inspect.

38. user (2026-06-21T17:44:31.718Z)

```
=== check for any [tool.hatch.build.targets.wheel] force-include / artifacts / only-include ===  
[tool.hatch.metadata]
```

The `reporting` extra pins obsreport to a git tag (a PEP 508 *direct reference*); hatchling requires opting in to allow those in metadata.

```
allow-direct-references = true
```

```
[tool.hatch.build.targets.wheel]
```

`backend` is a namespace package (no `\_\_init\_\_.py`); list both import roots explicitly so an editable/wheel install exposes `import backend` and `import training` regardless of cwd.

```
packages = ["backend", "training"]
```

39. assistant (2026-06-21T17:44:37.261Z)

Now let me actually build the wheel and confirm `backend/ui/` data files land in it. This is the load-bearing packaging claim (already-verified per the prompt, but I should reproduce adversarially). Critical concern: hatchling, when `packages` is set, by default includes only files matching the package patterns AND respects `.gitignore`/VCS ? but the UI files are **untracked** (`??` in git status). Hatchling's default file selection uses VCS (git) to determine what to include, and **untracked files may be excluded**. This could mean the wheel does NOT include `backend/ui/` until the files are committed. Let me test both ways.

40. user (2026-06-21T17:44:45.317Z)

```
=== build wheel (untracked backend/ui state) ===
```

```
* Creating isolated environment: venv+pip...
```

```
* Installing packages in isolated environment:
```

```
- hatchling
```

```
* Getting build dependencies for wheel...
```

```
* Building wheel...
```

```
Successfully built badminton_highlight_indexer-0.1.0-py3-none-any.whl
```

```
=== inspect wheel for backend/ui files ===
```

```
wheel: dist_audit/badminton_highlight_indexer-0.1.0-py3-none-any.whl
```

```
UI FILES IN WHEEL: 10
```

```
backend/ui/index.html
```

```
backend/ui/ui_version.json
```

```
backend/ui/assets/index-FJh_kElx.css
```

```
backend/ui/assets/index-ZjpoX6Fg.js
```

```
backend/ui/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
```

```
backend/ui/assets/noto-sans-devanagari-latin-ext-wght-normal-CLkupaxV.woff2
```

```
backend/ui/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
```

```
backend/ui/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
```

```
backend/ui/assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
```

```
backend/ui/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
```

41. assistant (2026-06-21T17:44:49.041Z)

All 10 `backend/ui/` files (including the woff2 fonts) land in the wheel even while untracked. The packaging claim holds. Let me clean up the audit artifact and do a final check on the licensing concern ? whether OFL fonts / minified React are flagged anywhere, and whether the python license-scan lane covers JS.

42. user (2026-06-21T17:44:54.413Z)

```
=== license-scan lane: does it scan JS? ===
.github/workflows/ci.yml
---
139:      - name: Install runtime deps + pip-licenses
142:        python -m pip install pip-licenses

=== any LICENSE/NOTICE bundled with the UI? ===

=== does khelsutra origin remote in ui_version.json suggest a private repo? ===
"spa_repo": "https://github.com/avidullu/khelsutra.git",
```

43. assistant (2026-06-21T17:45:02.774Z)

Confirmed: the license-scan lane (ci.yml:142) uses `pip-licenses` ? a **Python-only** scanner. It does NOT see the minified third-party JS (React etc.) baked into `backend/ui/assets/index-\*.js` or the OFL Noto fonts. No LICENSE/NOTICE is vendored alongside the bundle. This is a legitimate gap to NOTE given the project's hard licensing guardrails (MIT/Apache/BSD only for all deps including weights). The fonts are OFL (a permissive but copyleft-ish license requiring attribution/RFN compliance) and React is MIT ? all compatible, but the obligation (e.g. OFL requires the license to ship with the font, MIT requires the copyright notice) isn't satisfied by silently committing minified blobs with no NOTICE. Not a blocker for serving correctness, but the PR should note it.

I have enough evidence for all 7 review areas. Let me do one final confirmation of the broader test suite claim isn't broken by checking nothing else mounts `` or adds a catch-all in imported routers (the job_worker re-export, auth module).

44. user (2026-06-21T17:45:05.272Z)

```
C:\Users\avidu\Projects\bhi-wt-p1b\backend\api\uploads.py:20:from fastapi import APIRouter,
HTTPException, Request
C:\Users\avidu\Projects\bhi-wt-p1b\backend\api\uploads.py:26:router = APIRouter()
C:\Users\avidu\Projects\bhi-wt-p1b\backend\api\uploads.py:84:         _uploads[upload_id] =
{"filename": name, "tmp_path": tmp_path, "parts": 0, "bytes": 0}
C:\Users\avidu\Projects\bhi-wt-p1b\backend\api\uploads.py:125:         """Finalize: verify part
count, move into place, return {path, video_id} for /api/process."""
C:\Users\avidu\Projects\bhi-wt-p1b\backend\config\loader.py:128:
logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial data.")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:136:app.mount("/static",
StaticFiles(directory=_STATIC_DIR), name="static")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:147:     app.mount("/assets",
StaticFiles(directory=_UI_ASSETS_DIR), name="assets")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:150:@app.get("/",
response_class=HTMLResponse)
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:401:     logger.info(f"Running validations for
{req.video_path}...")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:528:         raise
HTTPException(status_code=404, detail=f"Video file not found at '{req.video_path}'")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:596:# backend.api.uploads as a self-contained
APIRouter. It resolves the live startup config via
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:601:app.include_router(uploads_api.router)
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:702:         return {"status": "success",
"filepath": out_path, "download_url": download_url}
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:710:     print(f"DEBUGGING VIDEO CLIP:
{video_path} at {timestamp}s")
```

```

C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:825:         print(f"[INFO] Invoking
diagnostics: {doctor_path} ...")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:846:         print(f"[CLI] Extracting debug
segment from {args.trim_start}s to {args.trim_end}s from {args.video_path}...")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:856:         print(f"[SUCCESS] Trimmed
segment saved to: {res_path}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:899:         print(f"[FATAL] Another
instance is already using this database ({db_path}). Refusing to "
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:914:         print(f"[CLI] Processing video
file: {args.video_path}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:925:         print(f"[FAIL] Video file not
found: {args.video_path}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:976:         print(f"[SUCCESS] Process
complete. Indexed {len(segments)} play segments inside database: {db_path}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:988:         print(f"[REPORT]
Technical report : {paths.get('technical_md')}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:989:         print(f"[REPORT] Customer
summary : {paths.get('customer_md')}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\main.py:990:         print(f"[REPORT] Machine
JSON      : {paths.get('json')}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\tools\rally_slicer.py:123:         raise
FileNotFoundError(f"video not found: {video_path}")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\tools\rally_slicer.py:154:
print(f"[slicer] rally {w.rally} -> {out_path} [{w.start:.3f}-{w.end:.3f}s]")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\pipeline\detectors\base.py:121:         res =
self._wsl_bash(f"df -P -B1 {wsl_tilde_quote(wsl_path)} | awk 'NR==2{{print $4}}'")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\pipeline\stitcher\compiler.py:137:
logger.warning(f"[watermark] logo not found: {logo_path} ? skipping logo overlay.")
C:\Users\avidu\Projects\bhi-wt-p1b\backend\pipeline\stitcher\compiler.py:173:
font_opt = f"fontfile='{self._ff_quote(font_path)}'"
C:\Users\avidu\Projects\bhi-wt-p1b\backend\pipeline\stitcher\compiler.py:180:
f"drawtext=textfile='{self._ff_quote(tf_path)}':{font_opt}"

```

[Showing results with pagination = limit: 30]

45. assistant (2026-06-21T17:45:08.497Z)

`uploads.py` defines an `APIRouter` included at line 601. Let me confirm its routes are all `/api`-prefixed (no catch-all that could shadow `/` or `/assets`).

46. user (2026-06-21T17:45:09.528Z)

```

64:@router.post("/api/upload/init")
89:@router.put("/api/upload/{upload_id}/part/{index}")
123:@router.post("/api/upload/{upload_id}/complete")
147:@router.delete("/api/upload/{upload_id}")

```

47. assistant (2026-06-21T17:45:45.212Z)

All uploads routes are `/api/upload/\*` prefixed ? no catch-all, no shadowing of `/` or `/assets`. The `{upload\_id}` and `{index}` path params are scoped under `/api/upload/`, so they cannot intercept SPA asset requests.

I now have complete file:line evidence for all 7 review areas. Let me compile the findings.

Adversarial review ? PR P1b (bundle khelsutra SPA into engine wheel, single-origin)

****[nit] backend/main.py:67 / scripts/bundle_ui.sh:67 ? `content\_hash` is path-dependent, not a reproducible content fingerprint** ? `CONTENT\_HASH=\$(find "\$UI\_DEST" ... | xargs sha256sum | sha256sum ...)` runs `find` over the *absolute* `UI\_DEST` path, and the outer `sha256sum` hashes a stream that embeds those absolute paths (verified: it reproduces `fad6253?` only with the literal `/c/Users/avidu/Projects/bhi-wt-p1b/...` prefix; a relative-path recompute gives**

`ac2a3d6?`). So re-bundling identical bytes from any other checkout path/machine/CI yields a different hash. It's recorded as provenance but isn't a verifiable fingerprint, and no test asserts it. ? Fix: hash with paths relativized to `\$UI\_DEST` (e.g. `cd "\$UI\_DEST" && find . -type f ! -name ui\_version.json | LC\_ALL=C sort | xargs sha256sum | sha256sum`), or drop the field if it's only decorative.

[nit] scripts/bundle_ui.sh:66 ? `API\_VERSION` extraction is brittle to a type annotation ? the anchor `^API\_VERSION\s\*=` matches today's `API\_VERSION = 1` (verified ? `1`), but a future `API\_VERSION: int = 1` would not match ? `ENGINE\_API\_VERSION` empty ? json written as `engine\_api\_version: null`. This fails **safe** (the skew test then asserts `null == 1` and fails loudly, prompting a re-bundle), so it's not a guard hole ? just a footgun. ? Fix: loosen to `^API\_VERSION\b[^\s]\*=` or note the assumption.

[minor] backend/ui/assets/*.js + *.woff2 ? minified third-party JS (React etc.) + OFL Noto fonts are committed into the MIT engine repo, and the license-scan lane does not cover them ? `.github/workflows/ci.yml:142` runs `pip-licenses` (Python-only); it cannot see the JS/font blobs. No `LICENSE`/`NOTICE` is vendored alongside the bundle (verified: none under `backend/ui/`). Given the project's hard MIT/Apache/BSD-only guardrail for *all* deps, and that OFL/MIT carry attribution obligations, this warrants an explicit note in the PR (and ideally a vendored third-party-notices file from the SPA build). Not a serving/correctness blocker. ? Fix: PR should NOTE the JS/font licensing posture; consider vendoring the SPA's generated licenses file into `backend/ui/`.`

Verified clean (no issue):

- **Serving correctness** ? `serve\_index` (main.py:151) reads `\_UI\_DIR/index.html`; `/assets` mount (main.py:147) is guarded by `os.path.isdir(\_UI\_ASSETS\_DIR)` (146), registered at import only when the dir exists. No `/assets`-prefixed API route exists ? no conflict; `/static` (136) is a distinct prefix. Fallback when `backend/ui` absent: `\_UI\_DIR` ? `\_STATIC\_DIR` (144), legacy `index.html` references `/static/\*` served by the always-present `/static` mount (verified functional, no crash); `\_UI\_ASSETS\_DIR` becomes a non-existent `static/assets` so `/assets` is simply not mounted. Traversal: Starlette 1.0.0 `StaticFiles` normalizes and 404s `../` escapes.
- **Route order / shadowing** ? no `{path:path}` catch-all anywhere (main.py + `backend/api/uploads.py` router routes are all `/api/upload/\*`). Serving `index.html` at `/` (150) does not intercept `/api/\*` or `/static`.
- **Bundler robustness** ? `set -euo pipefail` (sh:17); option-3 uses `curl -fsSL` + `tar -xzf` into a `mktemp -d` with `trap ? EXIT` cleanup; `stage\_dist` validates `index.html` presence and does `rm -rf` + `mkdir -p` + `cp -r "\$dist"/` into a clean dest (correct). Untrusted-tarball traversal is the standard `tar` caveat but the source is the owner's own release artifact (acceptable). CI never bundles ? no `.github/` reference to `bundle_ui`/`npm`/`backend/ui` (verified); the snapshot is committed.`
- **Skew guard** ? genuine: `test\_spa\_engine\_contract\_version\_matches` (test_ui_bundle.py:46) loads the committed `ui\_version.json` and asserts `engine\_api\_version == m.API\_VERSION`; bumping `API\_VERSION` without re-bundling makes them differ ? FAIL. No spurious-pass path.
- **CRLF/binary** ? `.gitattributes:15 backend/ui/* -text` resolves to `text: unset` for .html/.js/.json/.woff2` (verified via `git check-attr`); repo has core.autocrlf=true` , so this rule is *necessary* and correct ? committed bytes will match Vite output. `ui_version.json` is also covered by -text` (it's generated, never hand-edited, so no LF normalization needed ? fine).`
- **/api/version robustness** ? `\_bundled\_ui\_version` (main.py:159) reads `\_BUNDLED\_UI\_DIR/ui\_version.json` and returns `None` on `(OSError, ValueError)` (covers missing file + parse error); `json` imported (main.py:4), `Optional/Dict/Any` imported (main.py:32). `/api/version` still 200s without a bundle.`
- **Wheel packaging** ? built the wheel from the current (untracked `backend/ui/`) tree: all 10 UI files including the woff2 fonts are present (`backend/ui/...`). No force-include needed; claim holds.
- **Targeted tests** ? `tests/test_ui_bundle.py` : 5 passed.`

One process flag (not a code issue): the entire change is uncommitted working-tree state on `feat/packaging-plb-bundle-ui` (HEAD is still P1a #300); backend/ui/` is untracked, .gitattributes`/`backend/main.py` modified-not-staged. All verification above reflects working-tree bytes ? re-confirm after git add` that the -text` attribute lands the bytes unmangled (it will, since .gitattributes` is in-tree). Did not re-run the full 1295-test suite / ruff / mypy / coverage (prompt states these were already verified green at 85.12%).`

Verdict: No blockers or majors ? serving, fallback, route-ordering, and skew-guard logic are

correct and well-guarded; address the JS/font licensing note (minor) and optionally the two bundler nits before merge.